

AICines | Product Analytics

FULL ANALYTICAL REPORT

Prepared for: AICines Data Analyst Application
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Analysis period: January 2026 – August 2026
Analytics stack: PostgreSQL • SQL • Power Query • Power BI • DAX

15K	10K	60K	132.26M	\$1.31M
USERS	CREATORS	CONTENT	VIEWS	REVENUE

CASE STUDY DISCLOSURE

This is a portfolio case study created for the AICines Data Analyst application. AICines' internal user, creator, content-performance, event and transaction data is not publicly available. Accordingly, the operational dataset used in this analysis is synthetic, designed from publicly observable AICines product characteristics and common product-analytics structures. It is not AICines internal data and the figures must not be interpreted as actual AICines business performance.

The report explains the analytical approach, data architecture, validation process, dashboard design, findings, recommendations and how the framework could be extended with production data.

1. Executive Summary

This case study demonstrates an end-to-end product analytics workflow designed specifically for an AICines Data Analyst application. The goal was to show how I would translate an AI-powered content product into measurable product, engagement, creator and monetization questions, then build a reliable analytical solution around those questions.

The analysis is organized around five areas:

- **User growth & acquisition** — understanding modeled user scale and acquisition mix.
- **Creator activity & rewards** — measuring creator participation, tiers, publication activity and reward coverage.
- **Content performance & engagement** — comparing views, interactions, watch time and content types/genres.
- **Monetization** — evaluating revenue, paying users, transactions, average transaction value and credit purchases.
- **Product activity** — using event data to understand creation, publication, recreation and remix behaviour.

The final Power BI dashboard contains four pages: **Executive Overview**, **Creator Analytics**, **Content & Engagement**, and **Monetization**.

2. Product Context & Analytical Framing

Publicly observable AICines product information was used to understand the product context for this exercise. The platform is positioned around AI-powered content creation, video generation, editing, hosting, monetization and creator-oriented experiences.

Those public characteristics were used to frame the analytical questions. They were **not** used to infer confidential company performance, internal user counts, actual revenue or proprietary operational behaviour.

Analytical principle: public product information establishes the business context; the synthetic dataset demonstrates the analytical method.

3. Data Source & Synthetic Dataset

The required internal operational data was not publicly available. Rather than imply access to proprietary AICines information, a controlled synthetic dataset was created for this portfolio exercise.

The synthetic model was designed around common product-analytics entities that would be relevant to an AI content platform.

Table	Rows	Analytical role
dim_date	243	Calendar and time analysis
dim_users	15,000	User attributes and acquisition
dim_creators	10,000	Creator attributes and tiers
dim_content	60,000	Content attributes
fact_content_performance	60,000	Content performance snapshot
fact_user_events	220,000	Creation, publication and interaction events
fact_transactions	18,000	Purchases and revenue
fact_creator_rewards	16,000	Creator reward events

Total modeled records: **399,243**. All operational figures are synthetic.

4. Why Synthetic Data Was Introduced

- It allowed the project to model realistic relationships between users, creators, content and business events.
- It enabled a complete PostgreSQL → Power BI workflow.
- It provided enough volume for meaningful dashboard interactions and segmentation.
- It allowed data-quality rules and relationships to be controlled and independently validated.
- It made the distinction between **public product knowledge** and **simulated operational data** explicit.

5. Data Architecture & SQL Preparation

PostgreSQL was used as the data layer, with a dimensional structure separating descriptive dimensions from business-process facts.

Dimensions

- dim_date
- dim_users
- dim_creators
- dim_content

Facts

- fact_user_events
- fact_transactions
- fact_creator_rewards
- fact_content_performance

SQL was used to load, structure, validate and analyze the data. Analytical views were created for acquisition, content, creator, genre, monthly product and monthly revenue analysis.

6. Data Quality & Validation

Data quality was treated as part of the analytical workflow rather than an afterthought. The synthetic generation process controlled keys, categories, dates and relationships, while SQL and Power BI checks were used to validate the final model.

Validation checks included:

- Row counts across all analytical tables.
- Date-range and event-coverage checks.
- User and creator coverage.
- Transaction and revenue aggregation.
- Creator reward-event totals.
- Content-performance totals.
- Power BI relationship and filter-path testing.
- DAX measure validation.
- Blank-category cleanup.
- Chronological month sorting.

Why were there no significant errors? Because the dataset was generated under controlled conditions, referential relationships and categorical values could be kept internally consistent. This was then checked through SQL and Power BI. This should not be confused with production data quality: a live environment would require continuous monitoring for duplicates, missing events, bots, anomalies, instrumentation gaps, late-arriving data and schema changes.

7. KPI Framework

User metrics

- Total Users
- Active Users
- Users Who Created
- Users Who Published
- Publish Rate

Creator metrics

- Total Creators
- Active Creators
- Creator Tier Distribution
- Creator Reward Coverage
- Reward Credits
- Publications by Creator Tier

Content metrics

- Total Content
- Total Views
- Likes, Comments, Recreates & Shares
- Engagement Rate
- Watch Minutes
- Average Watch Seconds
- Views per Content

Monetization metrics

- Total Revenue
- Paying Users
- Total Transactions
- Average Transaction Value
- Revenue per Paying User
- Credits Purchased

8. Dashboard Structure & Visual Story

Page 1 — Executive Overview

- KPI cards for users, creators, content, views and revenue.
- Monthly revenue trend and creator/content activity.
- Content performance by genre.
- Acquisition source performance.

Page 2 — Creator Analytics

- Creator scale, active creators, reward coverage, published users and reward credits.
- Active creator trend, creator-tier distribution, reward credits by type and publications by creator tier.
- Filters: Date Range and Creator Tier.

Page 3 — Content & Engagement

- Content scale, views, engagement rate, watch minutes and views per content.
- Content interaction trend and content-type comparisons.
- Filters: Date Range and Content Type.

Page 4 — Monetization

- Revenue, paying users, transactions, average transaction value and revenue per paying user.
- Revenue vs Paying Users, Revenue by Acquisition Source, Average Transaction Value Trend and Credits Purchased Trend.
- Filters: Date Range and Acquisition Source.

9. Key Findings

The findings below are synthetic-model findings intended to demonstrate analytical interpretation. They are not statements about AICines' actual internal performance.

Platform scale

- 15,000 modeled users, 10,000 modeled creators and 60,000 modeled content records.
- The content-performance snapshot contains 132.26M modeled views.

Engagement

- Modeled overall engagement rate is 8.43%, calculated from likes, comments, recreates and shares relative to views.
- Content types show relatively close engagement-rate values, so volume and deeper engagement should be considered together.

Creator ecosystem

- Creator Reward Coverage is 75.8% in the modeled dataset.
- Emerging creators represent the largest modeled tier, followed by Growing, Established and Top Creator.

Monetization

- Modeled revenue is \$1.31M across 18K transactions.
- 10,514 modeled paying users are represented in the transaction data.
- Average Transaction Value is \$72.77 and Revenue per Paying User is \$124.59.

10. Important Data Interpretation Decision

The content-performance table contains a single 31 August 2026 snapshot. Therefore, it was not used to manufacture a January–August historical views trend.

- Snapshot data is used for cross-sectional comparisons such as content type and genre.
- Transaction and event data with valid dates are used for time-series analysis.
- This prevents a single snapshot from being incorrectly attributed to earlier publication dates.

This is an example of treating data limitations as an analytical constraint rather than forcing a visually attractive but misleading chart.

11. Recommendations

Strengthen creator conversion & retention

- Track the funnel from first creation to publication.
- Identify creator drop-off points.
- Compare retention and output across creator tiers.

Improve content discovery

- Evaluate views together with engagement and watch time.
- Identify content characteristics associated with deeper consumption.
- Avoid optimizing discovery solely on impressions.

Improve monetization analytics

- Track revenue alongside paying-user growth.
- Compare acquisition channels by both volume and revenue contribution.
- Investigate which channels produce higher-value users.

Evaluate the credit economy

- Connect credit purchasing to creation and interaction behaviour.
- Segment purchasing behaviour by user and creator characteristics.
- Investigate conversion from free usage to paid consumption.

Build cohort, funnel & experimentation analytics

- Introduce user and creator retention cohorts.
- Measure acquisition → creation → publication → engagement → monetization.
- Use A/B testing to evaluate onboarding, discovery, creation and monetization changes.
- Develop churn, reactivation, revenue-cohort and lifetime-value metrics.

12. How I Would Approach This at AICines

With access to production data, I would evolve this dashboard from descriptive reporting into a decision-support framework tied directly to product outcomes.

- **Retention:** user and creator cohorts to measure sustained product value.
- **Activation:** identify behaviours that signal successful creator and user onboarding.
- **Content:** analyze drivers of watch time, engagement and repeat consumption.

- **Monetization:** connect user behaviour, credit consumption, transaction value and customer lifetime value.
- **Experimentation:** partner with Product and Growth teams to define measurable hypotheses and A/B tests.
- **Stakeholder reporting:** maintain consistent KPI definitions across Product, Growth, Creator and Finance teams.

The goal would be to move from “What happened?” toward “Why did it happen, what should we test, and did the intervention work?”

13. Limitations

- The operational dataset is synthetic and is not AICines internal data.
- Revenue, users, transactions, views, events and engagement values are simulated.
- The results cannot be used to describe AICines' actual financial or product performance.
- Content-performance data is a snapshot rather than a complete historical performance series.
- Production analytics would require source-system access, validated event instrumentation and agreed KPI definitions.

14. Conclusion

This case study demonstrates a complete product analytics workflow: understand the product context, define business questions, structure the data, prepare and validate it with SQL, develop reusable DAX measures, build an executive Power BI experience and translate analytical patterns into recommendations.

The project is deliberately transparent about its data limitations. Public AICines product information was used for context, while the operational layer was synthetically constructed for analytical demonstration. This allows the work to demonstrate practical data and product-analytics skills without implying access to confidential company information.

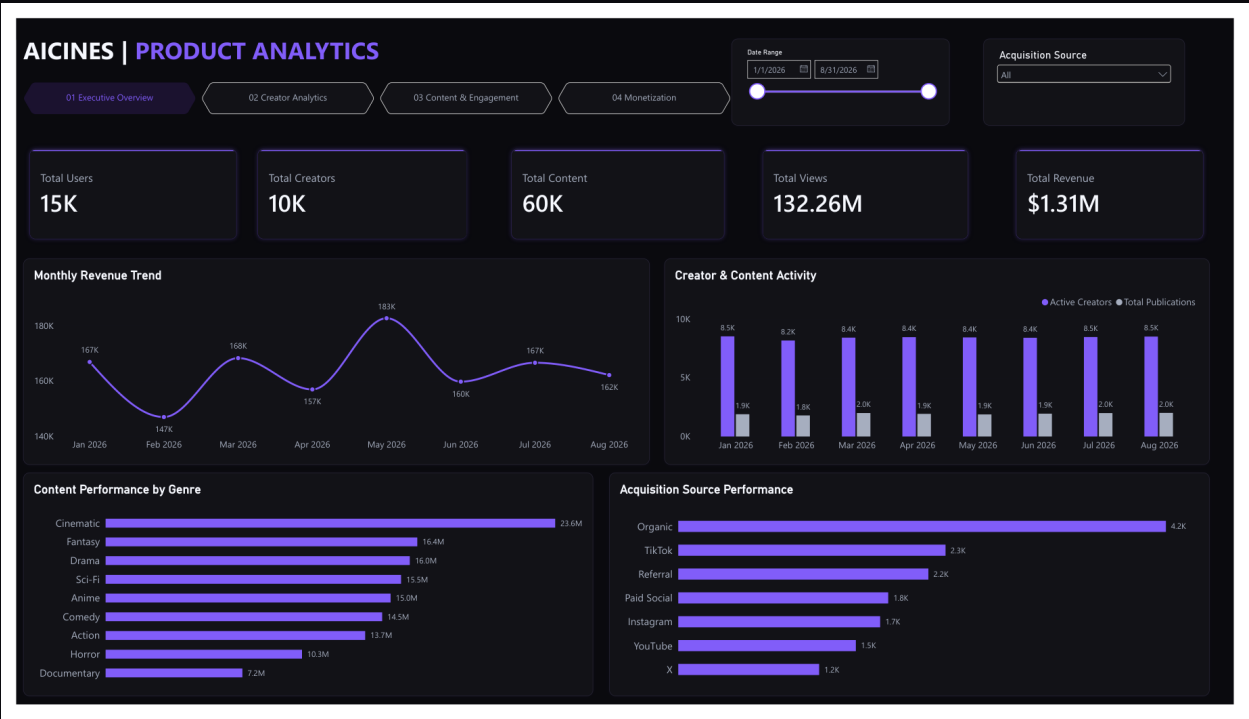
The next logical step in a real AICines environment would be to replace the synthetic layer with validated production data and extend the framework into retention, cohorts, funnels, experimentation, churn and lifetime-value analysis.

Overall analytical objective: turn product data into clear evidence that helps teams understand user behaviour, creator health, content performance and monetization — and use those insights to make measurable product decisions.

Appendix A — Dashboard Reference

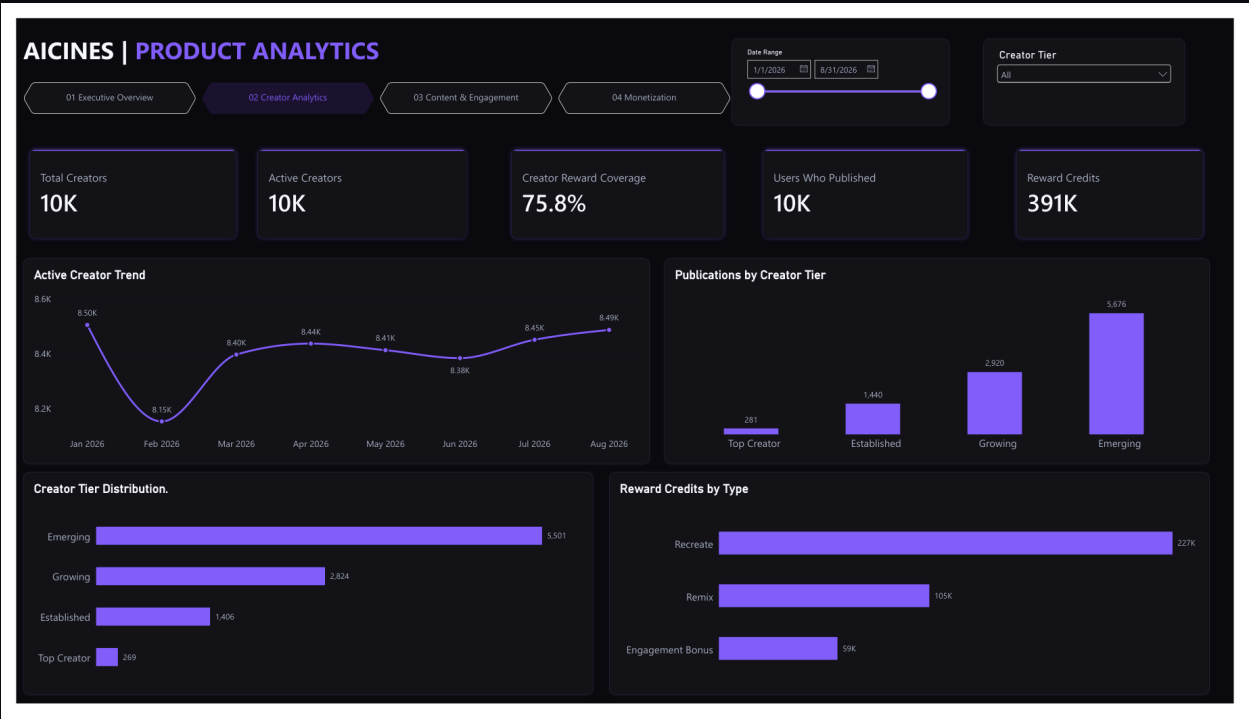
The submitted Power BI dashboard contains four pages. The thumbnails below reproduce the visual structure of the dashboard used for this case study.

Page 1 — Dashboard reference



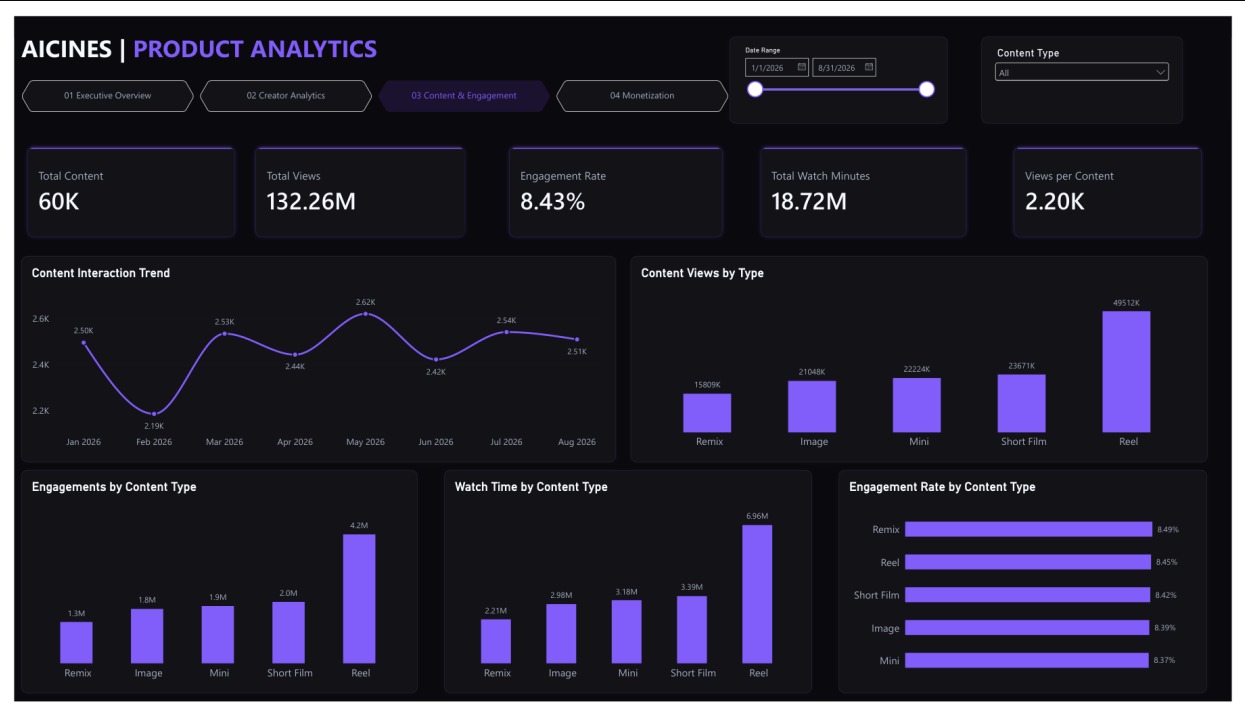
Appendix A — Dashboard Reference

Page 2 — Dashboard reference



Appendix A — Dashboard Reference

Page 3 — Dashboard reference



Appendix A — Dashboard Reference

Page 4 — Dashboard reference

